

REG NO:192372295

1. Weather-Based Play Prediction

Build a WEKA application that predicts whether a person should play outdoors based on weather conditions.

- Dataset: Outlook, Temperature, Humidity, Windy
- Target: Play (Yes/No)
- Algorithm: **J48 Decision Tree**
- Tasks:
 1. Load the dataset into WEKA.
 2. Preprocess the data.
 3. Apply J48.
 4. Visualize the decision tree.
 5. Predict Play for a new weather condition.

Step 1: Open WEKA Explorer

- Open WEKA GUI Chooser.
- Click Explorer.

Step 2: Load the Weather dataset

- Go to the Preprocess tab.
- Click Open file.
- Select weather.arff.
- Click Open.

The dataset contains:

- Attributes: Outlook, Temperature, Humidity, Windy, Play
- Class attribute: Play (Yes/No)

Step 3: Select the classification algorithm • Click the Classify tab.

- Click Choose.
- Select trees → J48.

The selected classifier becomes:

J48 -C 0.25 -M 2

Step 4: Choose the evaluation method

Under Test options:

- Select Use training set.

- (Or select Cross-validation if required.)

Step 5: Train the model

- Click Start.

WEKA builds the decision tree and evaluates the classifier. **Step 6:**

View the classifier output

The output displays:

- Correctly classified instances
- Incorrectly classified instances
- Accuracy
- Error values
- Confusion matrix

Step 7: Visualize the decision tree

- In the Result list, right-click trees.J48.
- Click Visualize tree.

The generated tree shows the decision rules.

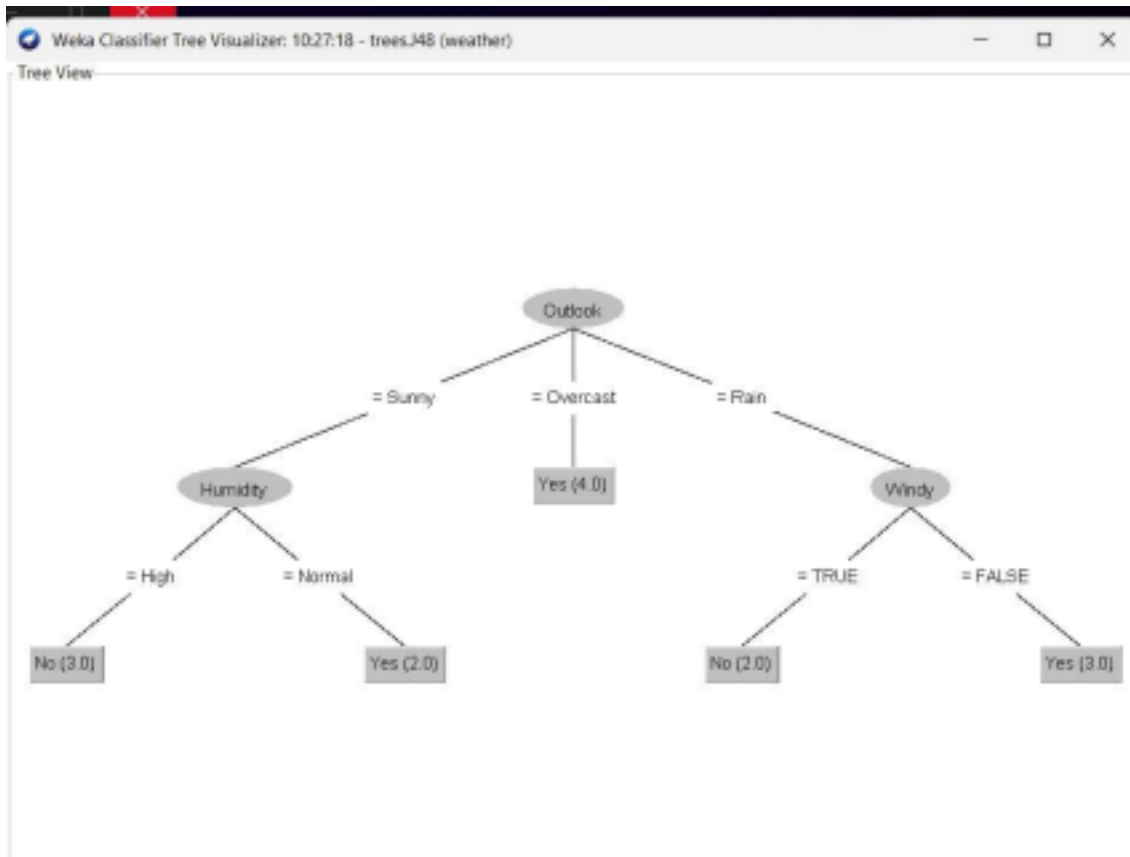
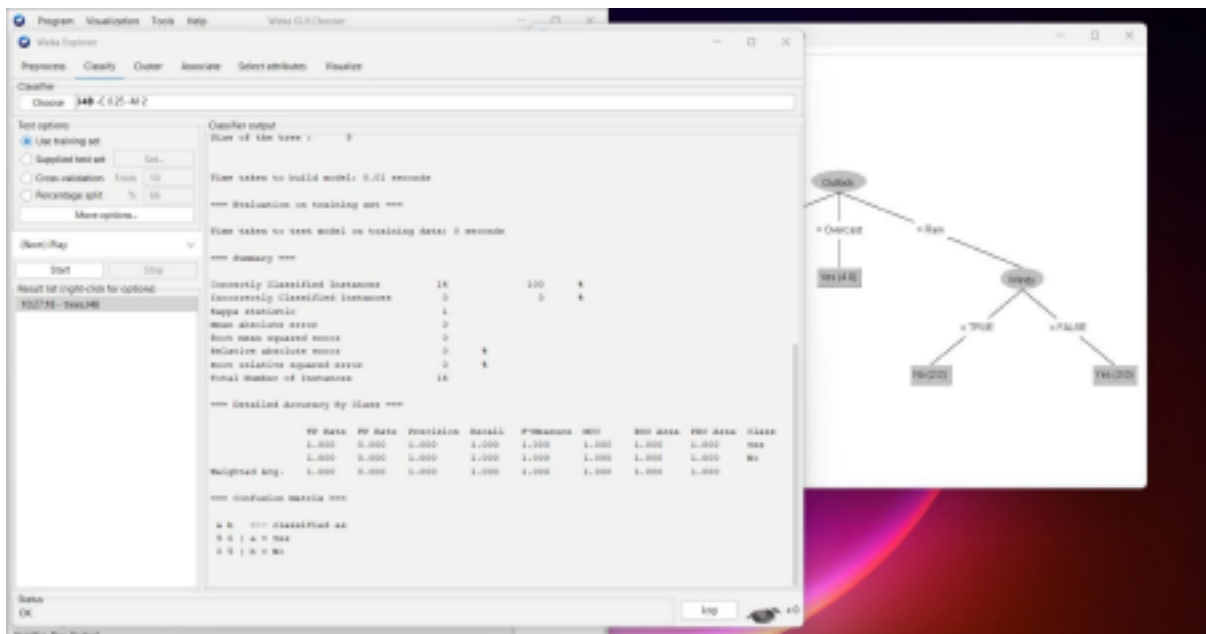
Step 8: Classify a new weather condition • Create weather_test.arff with a

- new weather record. • Select Supplied test set.
- Load the test file.
- Click Start.

The model predicts Play = Yes or No.

Result

- The weather dataset was successfully classified using the J48 algorithm. • The decision tree was generated successfully.
- The model predicted the play decision for new weather conditions.



2. Flower Species Classification

Build a WEKA application that identifies the species of a flower from its physical measurements.

- Dataset: Iris dataset
- Attributes: Sepal Length, Sepal Width, Petal Length, Petal Width • Target: Iris Species
- Algorithm: **J48 or Naive Bayes**
- Tasks:
 1. Load the Iris dataset.
 2. Select the classification algorithm.
 3. Train the model.
 4. Evaluate its accuracy.

5. Classify a new flower.

Step 1: Open WEKA Explorer

- Open WEKA GUI Chooser.
- Click Explorer.

Step 2: Load the Iris dataset

- Go to the Preprocess tab.
- Click Open file.
- Select iris.arff.
- Click Open.

The dataset contains:

- Instances: 110
- Attributes:
 - sepallength
 - sepalwidth
 - petallength
 - petalwidth
 - class

Step 3: Select the classification algorithm • Click the Classify tab.

- Click Choose.
- Select trees → J48. The selected classifier becomes:

J48 -C 0.25 -M 2

Step 4: Choose the evaluation method Under Test options:

- Select Cross-validation.
- Set Folds = 10.

Step 5: Train the model

- Click Start.

WEKA builds the decision tree and evaluates the classifier.

Step 6: Evaluate the model

The output displays:

- Correctly classified instances: 104
- Incorrectly classified instances: 6
- Accuracy: 94.5455%
- Confusion matrix

- Precision and recall values

Step 7: Visualize the data

- Click the Visualize tab.
- Observe the scatter plot matrix of all attributes.

The visualization shows the distribution of:

- Iris-setosa
- Iris-versicolor
- Iris-virginica

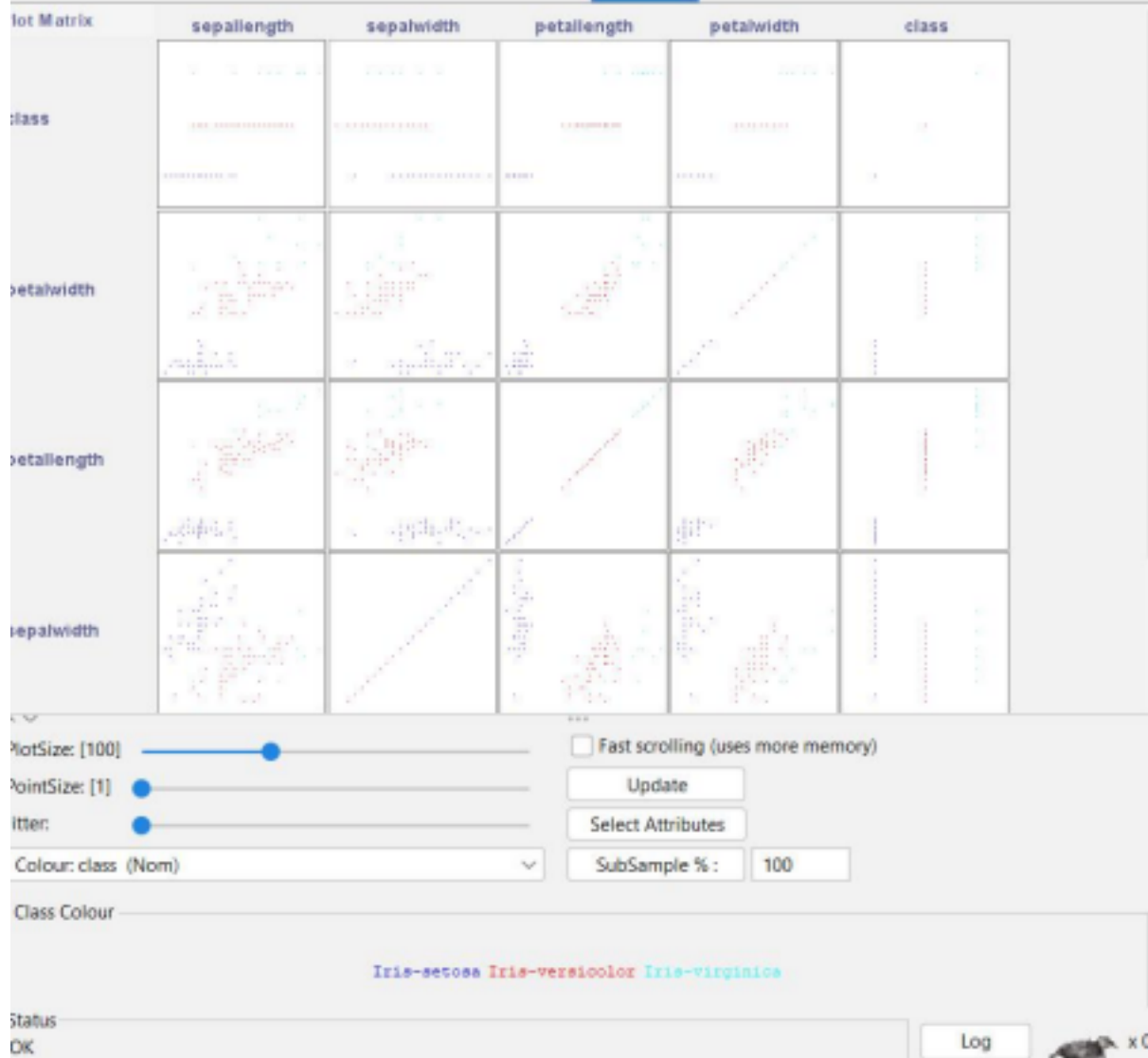
Step 8: Classify a new flower

- Create iris_test.arff with a new flower sample.
- Select Supplied test set.
- Load the test file.
- Click Start.

The model predicts the flower species.

Result

- The Iris dataset was successfully classified using the J48 algorithm.
- The model achieved 94.5455% accuracy.
- The classifier correctly identified most flower species.
- The trained model successfully predicted the species of a new flower sample.



Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier

Choose **J48 -C 0.25 -M 2**

Test options

☐ Use training set

☐ Supplied test set

☒ Cross-validation Folds

☐ Percentage split %

(Nom) class

Result list (right-click for options)

19.2252 - trees.J48

Classifier output

Size of the tree : 7

Time taken to build model: 0.01 seconds

=== Stratified cross-validation ===

=== Summary ===

Metric	Value	Percentage
Correctly Classified Instances	104	94.5455 %
Incorrectly Classified Instances	6	5.4545 %
Kappa statistic	0.9063	
Mean absolute error	0.0374	
Root mean squared error	0.1842	
Relative absolute error	9.6586 %	
Root relative squared error	41.9543 %	
Total Number of Instances	110	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MC
	0.980	0.000	1.000	0.980	0.990	0.
	0.940	0.050	0.940	0.940	0.940	0.
	0.800	0.030	0.727	0.800	0.762	0.
Weighted Avg.	0.945	0.025	0.948	0.945	0.946	0.

=== Confusion Matrix ===

	a	b	c	<-- classified as
49	1	0	0	a = Iris-setosa
0	47	3	0	b = Iris-versicolor
0	2	8	0	c = Iris-virginica

Status
OK

 x 0

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier

Choose **J48 -C 0.25 -M 2**

Test options

☐ Use training set

☒ Supplied test set

☐ Cross-validation Folds

☐ Percentage split %

(Nom) class

Result list (right-click for options)

19.2151 - trees.J48

19.2252 - trees.J48

Classifier output

=== Run information ===

Algorithm: weka.classifiers.trees.J48 -C 0.25 -M 2

Relation: iris

Instances: 110

Attributes: 5

sepalwidth
sepalwidth
petalwidth
petalwidth
class

Test mode: user supplied test set: size unknown (reading incrementally)

=== Classifier model (full training set) ===

J48 pruned tree

```

petalwidth <= 0.6: iris-setosa (50/0)
petalwidth > 0.6
  | petalwidth <= 1.6: iris-versicolor (46/0)
  | petalwidth > 1.6
    | petalwidth <= 5: iris-versicolor (2/0/1/0)
    | petalwidth > 5: iris-virginica (1/0/0)

```

Number of leaves : 4

Size of the tree : 7

Time taken to build model: 0 seconds

=== Evaluation on test set ===

Time taken to test model on supplied test set: 0 seconds

=== Summary ===

Status
OK